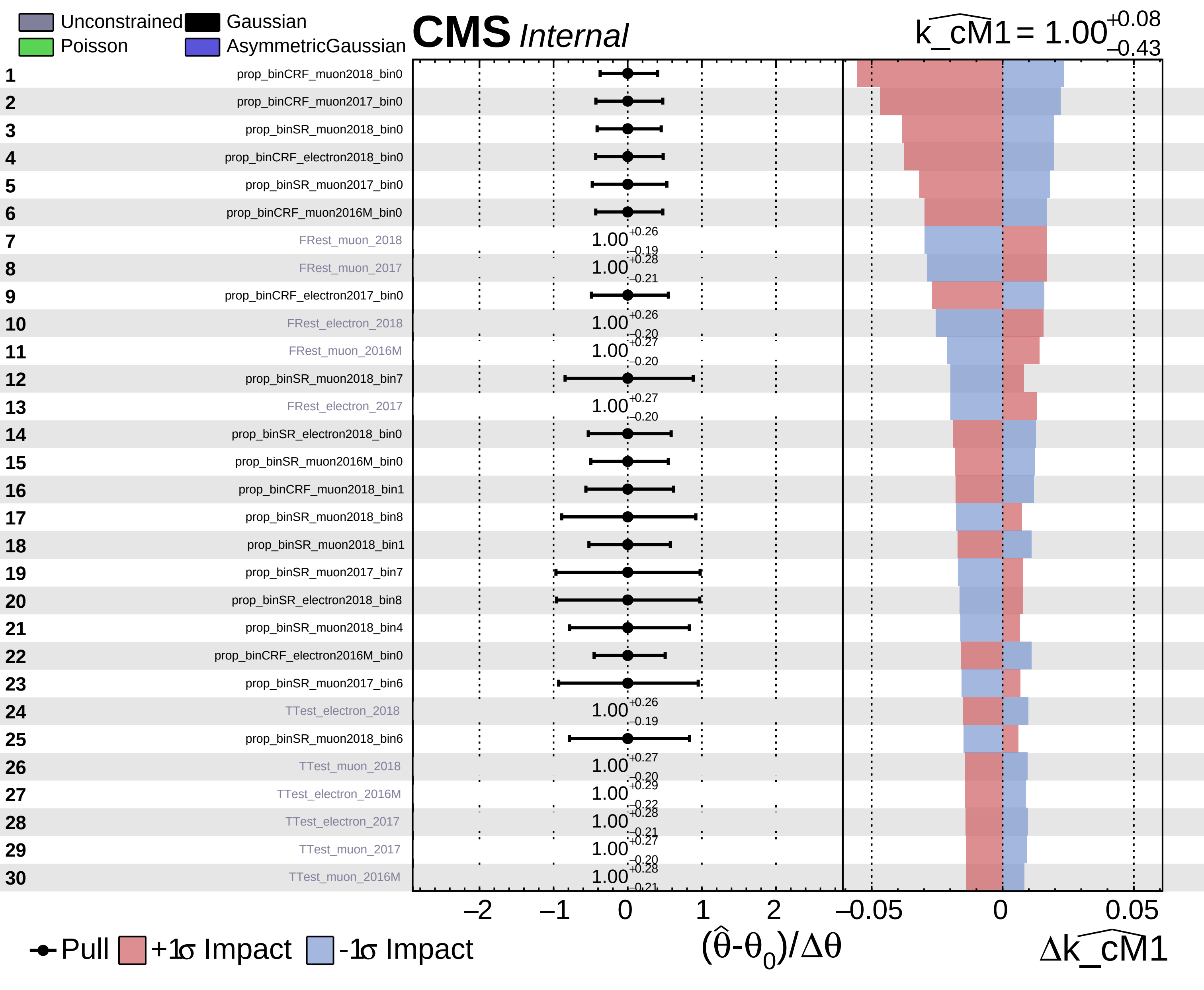
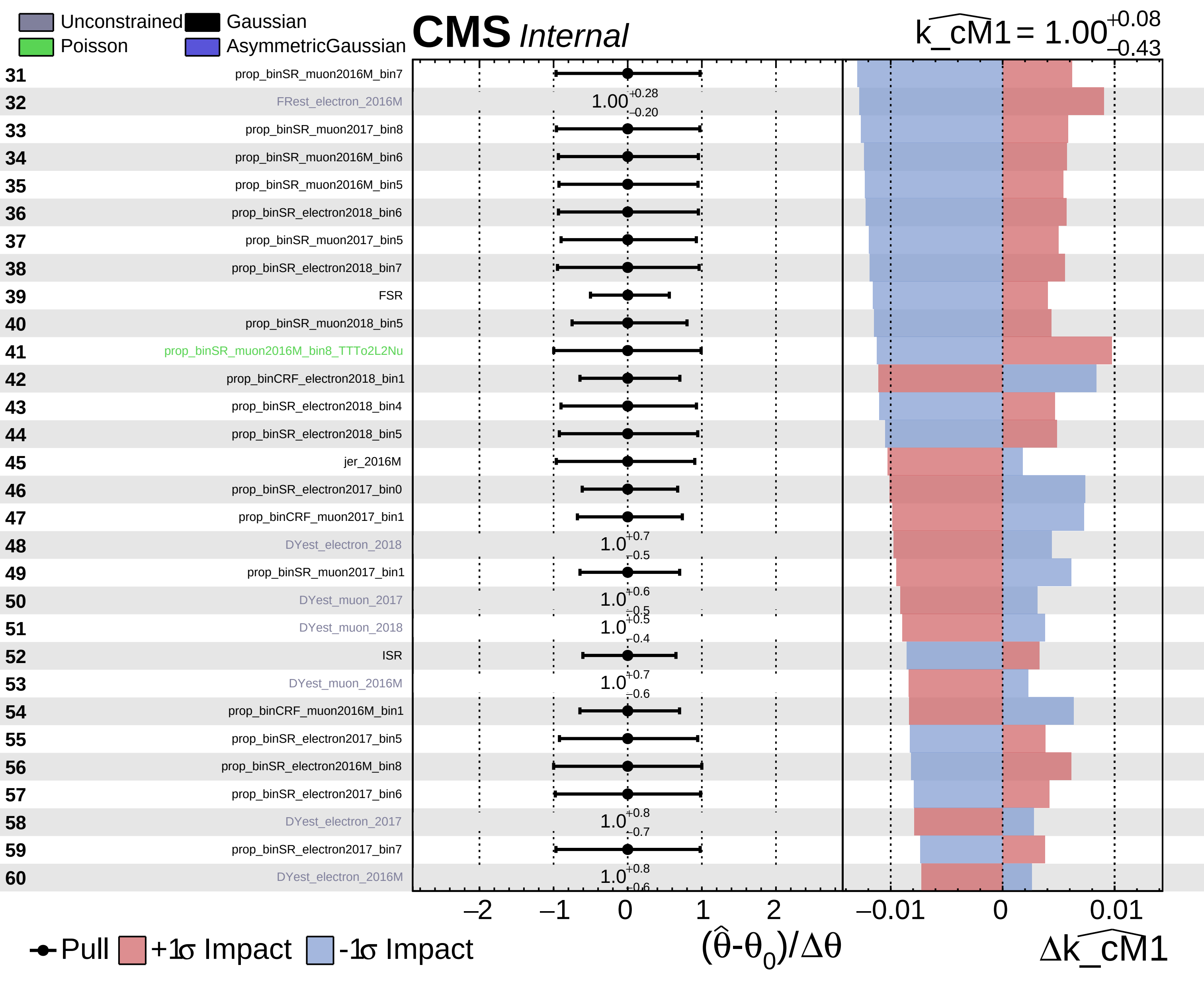

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
1 prop_binCRF_muon2018_bin0
2 prop_binCRF_muon2017_bin0
3 prop_binSR_muon2018_bin0
4 prop_binCRF_electron2018_bin0
5 prop_binSR_muon2017_bin0
6 prop_binCRF_muon2016M_bin0
7 FRest_muon_2018 1.00 +0.26 −0.19
8 FRest_muon_2017 1.00 +0.28 −0.21
9 prop_binCRF_electron2017_bin0
10 FRest_electron_2018 1.00 +0.26 −0.20
11 FRest_muon_2016M 1.00 +0.27 −0.20
12 prop_binSR_muon2018_bin7
13 FRest_electron_2017 1.00 +0.27 −0.20
14 prop_binSR_electron2018_bin0
15 prop_binSR_muon2016M_bin0
16 prop_binCRF_muon2018_bin1
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2018_bin1
19 prop_binSR_muon2017_bin7
20 prop_binSR_electron2018_bin8
21 prop_binSR_muon2018_bin4
22 prop_binCRF_electron2016M_bin0
23 prop_binSR_muon2017_bin6
24 TTest_electron_2018 1.00 +0.26 −0.19
25 prop_binSR_muon2018_bin6
26 TTest_muon_2018 1.00 +0.27 −0.20
27 TTest_electron_2016M 1.00 +0.29 −0.22
28 TTest_electron_2017 1.00 +0.28 −0.21
29 TTest_muon_2017 1.00 +0.27 −0.20
30 TTest_muon_2016M 1.00 +0.28 −0.21
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1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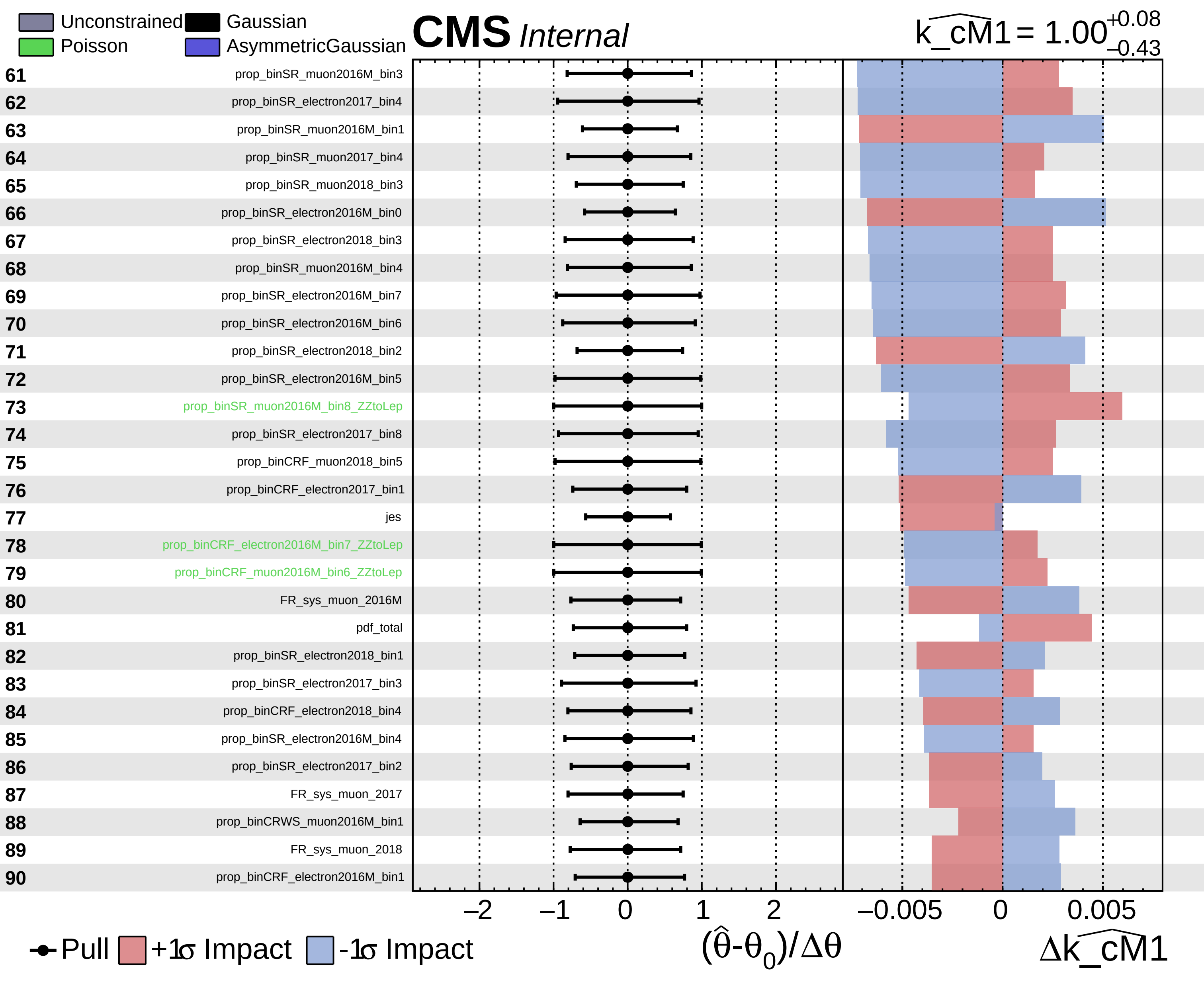
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
31 prop_binSR_muon2016M_bin7
32 FRest_electron_2016M 1.00 +0.28 −0.20
33 prop_binSR_muon2017_bin8
34 prop_binSR_muon2016M_bin6
35 prop_binSR_muon2016M_bin5
36 prop_binSR_electron2018_bin6
37 prop_binSR_muon2017_bin5
38 prop_binSR_electron2018_bin7
39 FSR
40 prop_binSR_muon2018_bin5
41 prop_binSR_muon2016M_bin8_TTTo2L2Nu
42 prop_binCRF_electron2018_bin1
43 prop_binSR_electron2018_bin4
44 prop_binSR_electron2018_bin5
45 jer_2016M
46 prop_binSR_electron2017_bin0
47 prop_binCRF_muon2017_bin1
48 DYest_electron_2018 1.0 +0.7 −0.5
49 prop_binSR_muon2017_bin1
50 DYest_muon_2017 1.0 +0.6 −0.5
51 DYest_muon_2018 1.0 +0.5 −0.4
52 ISR
53 DYest_muon_2016M 1.0 +0.7 −0.6
54 prop_binCRF_muon2016M_bin1
55 prop_binSR_electron2017_bin5
56 prop_binSR_electron2016M_bin8
57 prop_binSR_electron2017_bin6
58 DYest_electron_2017 1.0 +0.8 −0.7
59 prop_binSR_electron2017_bin7
60 DYest_electron_2016M 1.0 +0.8 −0.6
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1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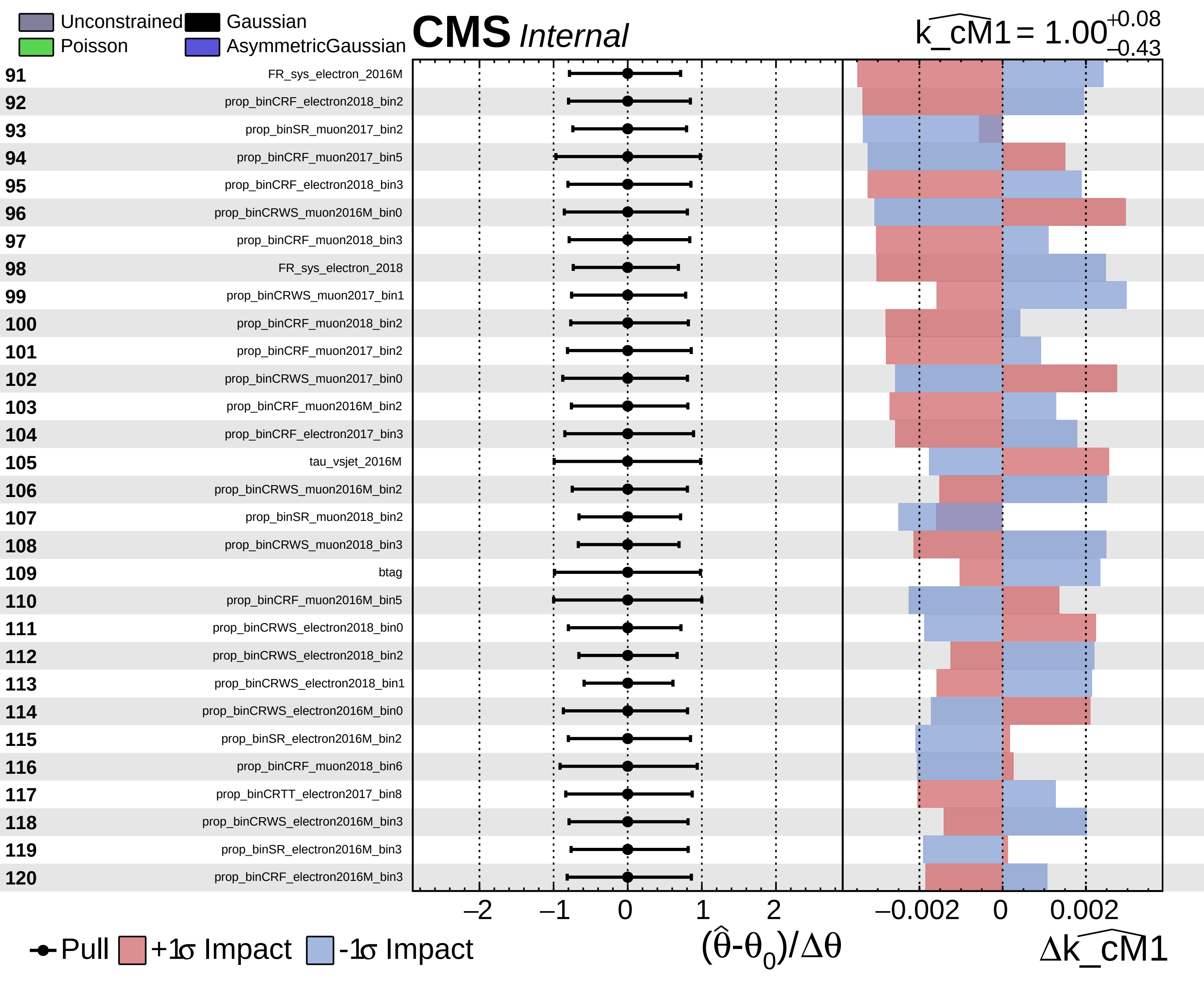
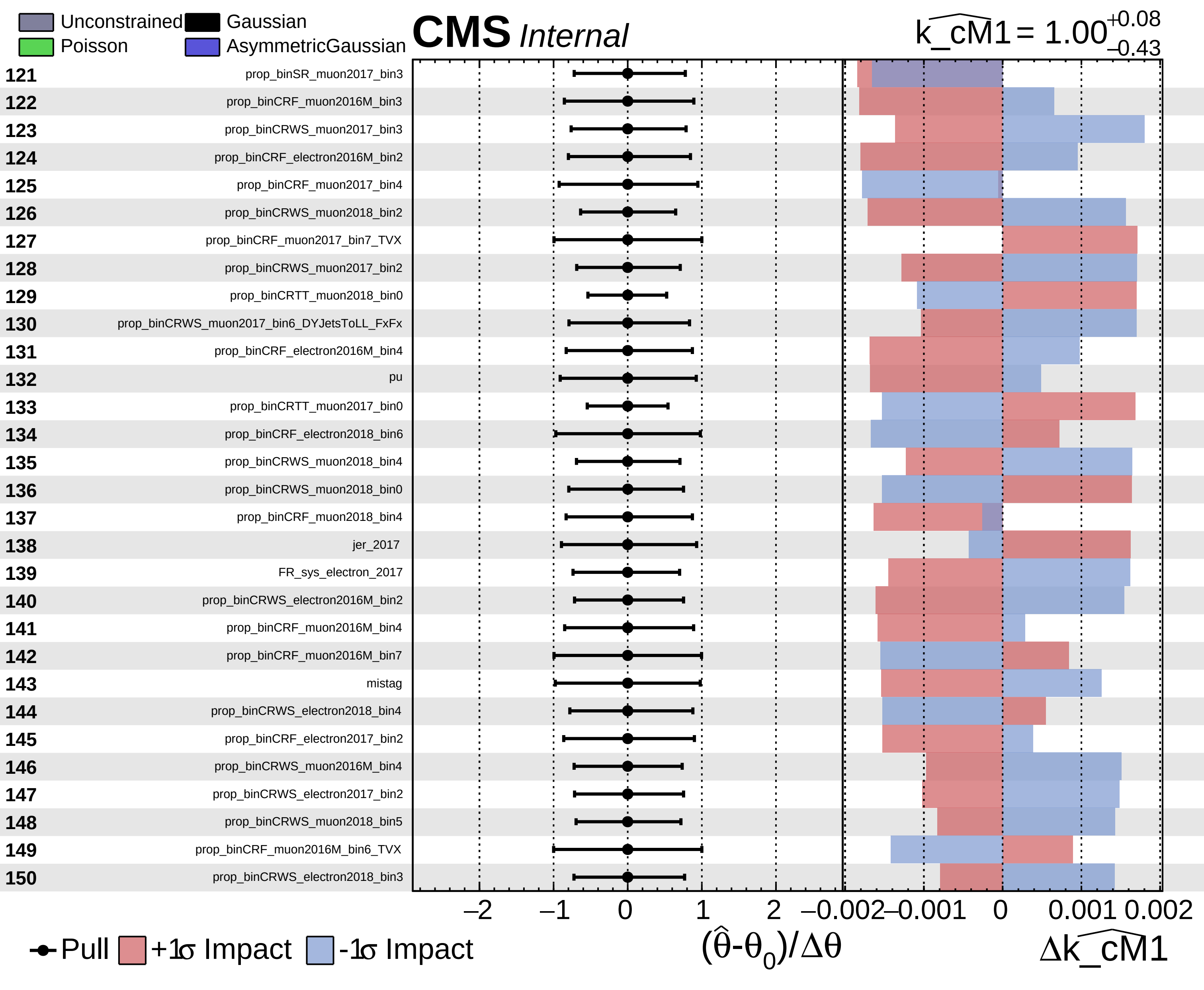
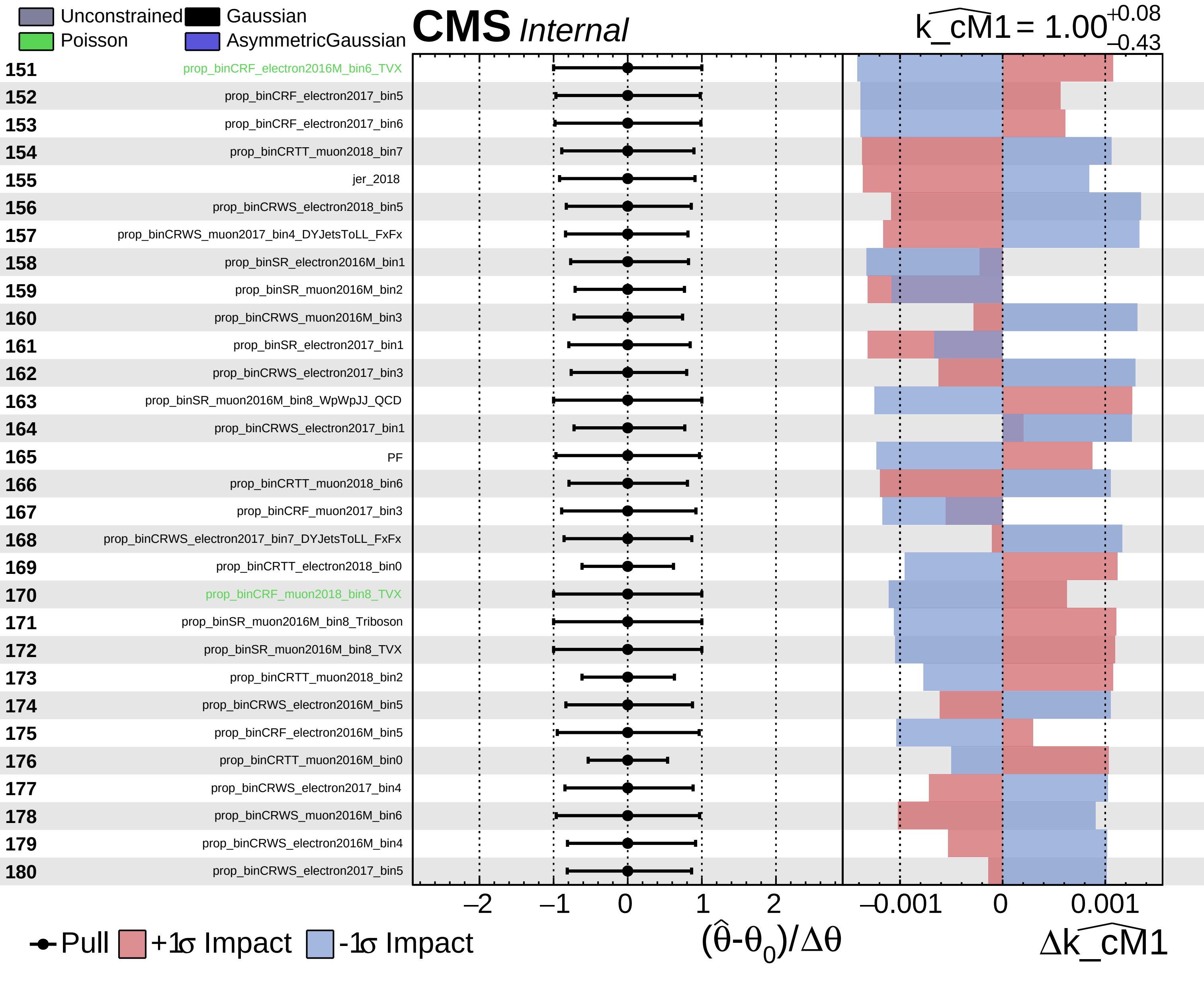
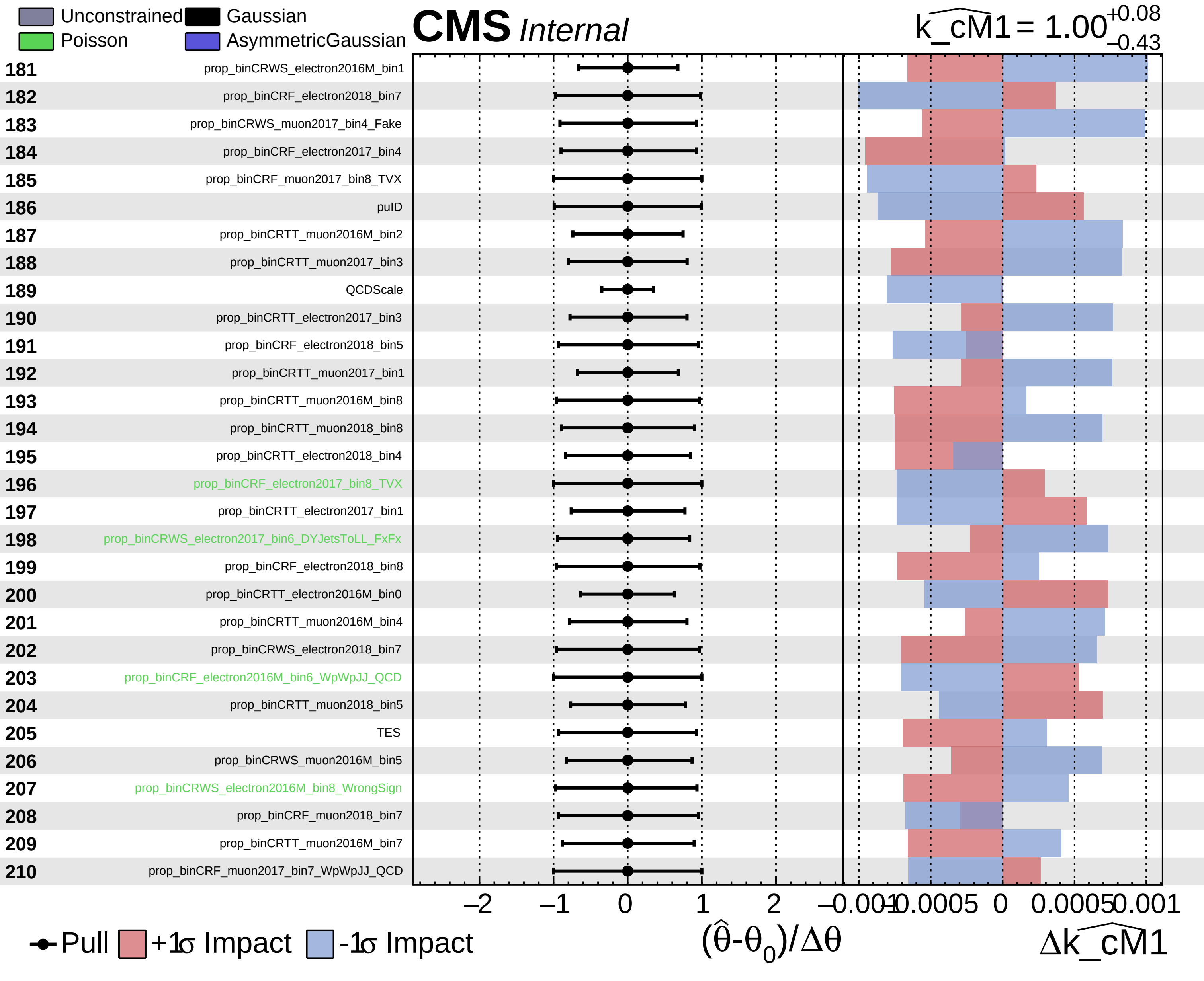
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 1.00^{+0.08}_{-0.43}$
61 prop_binSR_muon2016M_bin3
62 prop_binSR_electron2017_bin4
63 prop_binSR_muon2016M_bin1
64 prop_binSR_muon2017_bin4
65 prop_binSR_muon2018_bin3
66 prop_binSR_electron2016M_bin0
67 prop_binSR_electron2018_bin3
68 prop_binSR_muon2016M_bin4
69 prop_binSR_electron2016M_bin7
70 prop_binSR_electron2016M_bin6
71 prop_binSR_electron2018_bin2
72 prop_binSR_electron2016M_bin5
73 prop_binSR_muon2016M_bin8_ZZtoLep
74 prop_binSR_electron2017_bin8
75 prop_binCRF_muon2018_bin5
76 prop_binCRF_electron2017_bin1
77 jes
78 prop_binCRF_electron2016M_bin7_ZZtoLep
79 prop_binCRF_muon2016M_bin6_ZZtoLep
80 FR_sys_muon_2016M
81 pdf_total
82 prop_binSR_electron2018_bin1
83 prop_binSR_electron2017_bin3
84 prop_binCRF_electron2018_bin4
85 prop_binSR_electron2016M_bin4
86 prop_binSR_electron2017_bin2
87 FR_sys_muon_2017
88 prop_binCRWS_muon2016M_bin1
89 FR_sys_muon_2018
90 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$

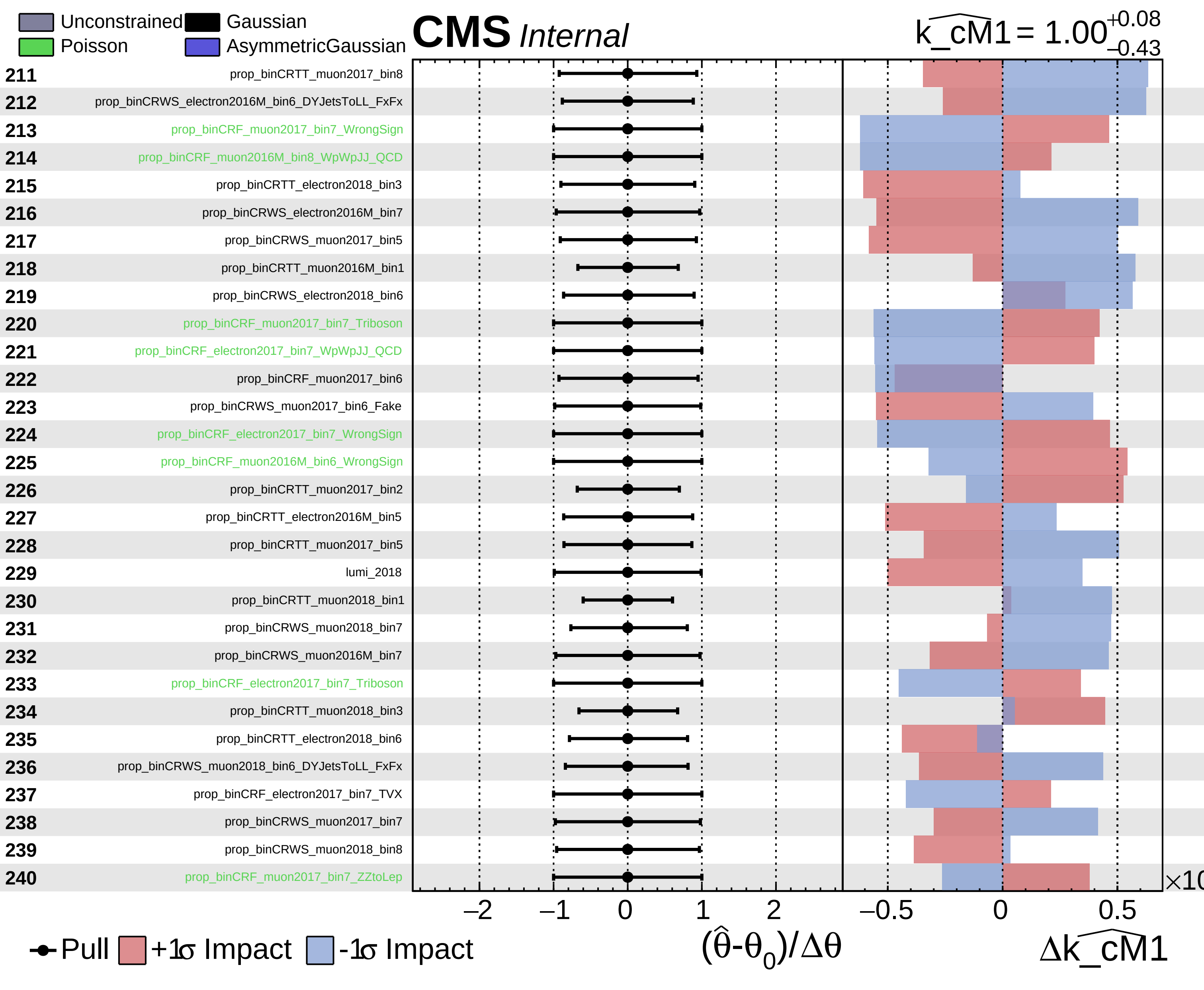

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
91 FR_sys_electron_2016M
92 prop_binCRF_electron2018_bin2
93 prop_binSR_muon2017_bin2
94 prop_binCRF_muon2017_bin5
95 prop_binCRF_electron2018_bin3
96 prop_binCRWS_muon2016M_bin0
97 prop_binCRF_muon2018_bin3
98 FR_sys_electron_2018
99 prop_binCRWS_muon2017_bin1
100 prop_binCRF_muon2018_bin2
101 prop_binCRF_muon2017_bin2
102 prop_binCRWS_muon2017_bin0
103 prop_binCRF_muon2016M_bin2
104 prop_binCRF_electron2017_bin3
105 tau_vsjet_2016M
106 prop_binCRWS_muon2016M_bin2
107 prop_binSR_muon2018_bin2
108 prop_binCRWS_muon2018_bin3
109 btag
110 prop_binCRF_muon2016M_bin5
111 prop_binCRWS_electron2018_bin0
112 prop_binCRWS_electron2018_bin2
113 prop_binCRWS_electron2018_bin1
114 prop_binCRWS_electron2016M_bin0
115 prop_binSR_electron2016M_bin2
116 prop_binCRF_muon2018_bin6
117 prop_binCRTT_electron2017_bin8
118 prop_binCRWS_electron2016M_bin3
119 prop_binSR_electron2016M_bin3
120 prop_binCRF_electron2016M_bin3
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
121 prop_binSR_muon2017_bin3
122 prop_binCRF_muon2016M_bin3
123 prop_binCRWS_muon2017_bin3
124 prop_binCRF_electron2016M_bin2
125 prop_binCRF_muon2017_bin4
126 prop_binCRWS_muon2018_bin2
127 prop_binCRF_muon2017_bin7_TVX
128 prop_binCRWS_muon2017_bin2
129 prop_binCRTT_muon2018_bin0
130 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
131 prop_binCRF_electron2016M_bin4
132 pu
133 prop_binCRTT_muon2017_bin0
134 prop_binCRF_electron2018_bin6
135 prop_binCRWS_muon2018_bin4
136 prop_binCRWS_muon2018_bin0
137 prop_binCRF_muon2018_bin4
138 jer_2017
139 FR_sys_electron_2017
140 prop_binCRWS_electron2016M_bin2
141 prop_binCRF_muon2016M_bin4
142 prop_binCRF_muon2016M_bin7
143 mistag
144 prop_binCRWS_electron2018_bin4
145 prop_binCRF_electron2017_bin2
146 prop_binCRWS_muon2016M_bin4
147 prop_binCRWS_electron2017_bin2
148 prop_binCRWS_muon2018_bin5
149 prop_binCRF_muon2016M_bin6_TVX
150 prop_binCRWS_electron2018_bin3
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1

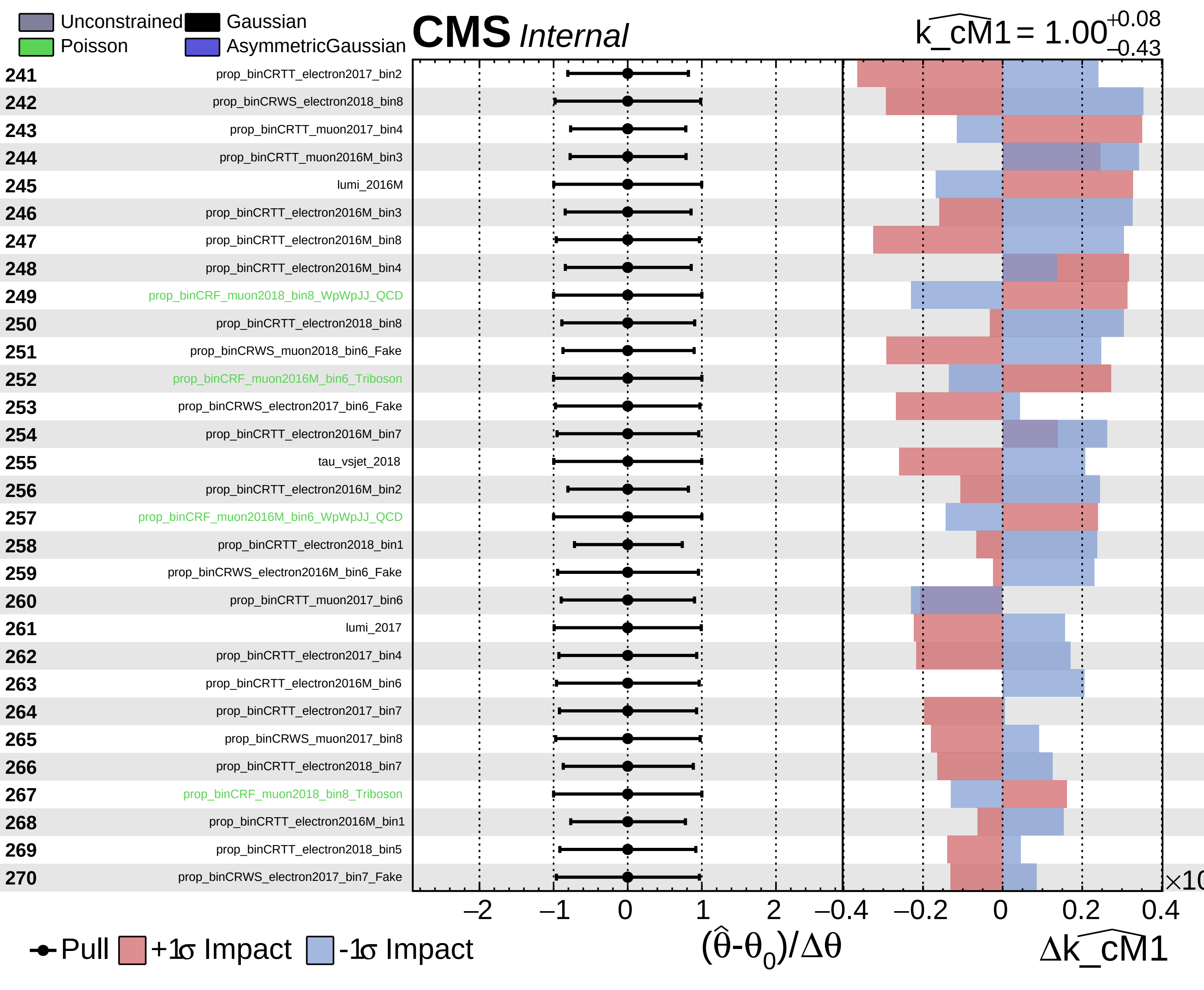


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
151 prop_binCRF_electron2016M_bin6_TVX
152 prop_binCRF_electron2017_bin5
153 prop_binCRF_electron2017_bin6
154 prop_binCRTT_muon2018_bin7
155 jer_2018
156 prop_binCRWS_electron2018_bin5
157 prop_binCRWS_muon2017_bin4_DYJetsToLL_FxFx
158 prop_binSR_electron2016M_bin1
159 prop_binSR_muon2016M_bin2
160 prop_binCRWS_muon2016M_bin3
161 prop_binSR_electron2017_bin1
162 prop_binCRWS_electron2017_bin3
163 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
164 prop_binCRWS_electron2017_bin1
165 PF
166 prop_binCRTT_muon2018_bin6
167 prop_binCRF_muon2017_bin3
168 prop_binCRWS_electron2017_bin7_DYJetsToLL_FxFx
169 prop_binCRTT_electron2018_bin0
170 prop_binCRF_muon2018_bin8_TVX
171 prop_binSR_muon2016M_bin8_Triboson
172 prop_binSR_muon2016M_bin8_TVX
173 prop_binCRTT_muon2018_bin2
174 prop_binCRWS_electron2016M_bin5
175 prop_binCRF_electron2016M_bin5
176 prop_binCRTT_muon2016M_bin0
177 prop_binCRWS_electron2017_bin4
178 prop_binCRWS_muon2016M_bin6
179 prop_binCRWS_electron2016M_bin4
180 prop_binCRWS_electron2017_bin5
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1

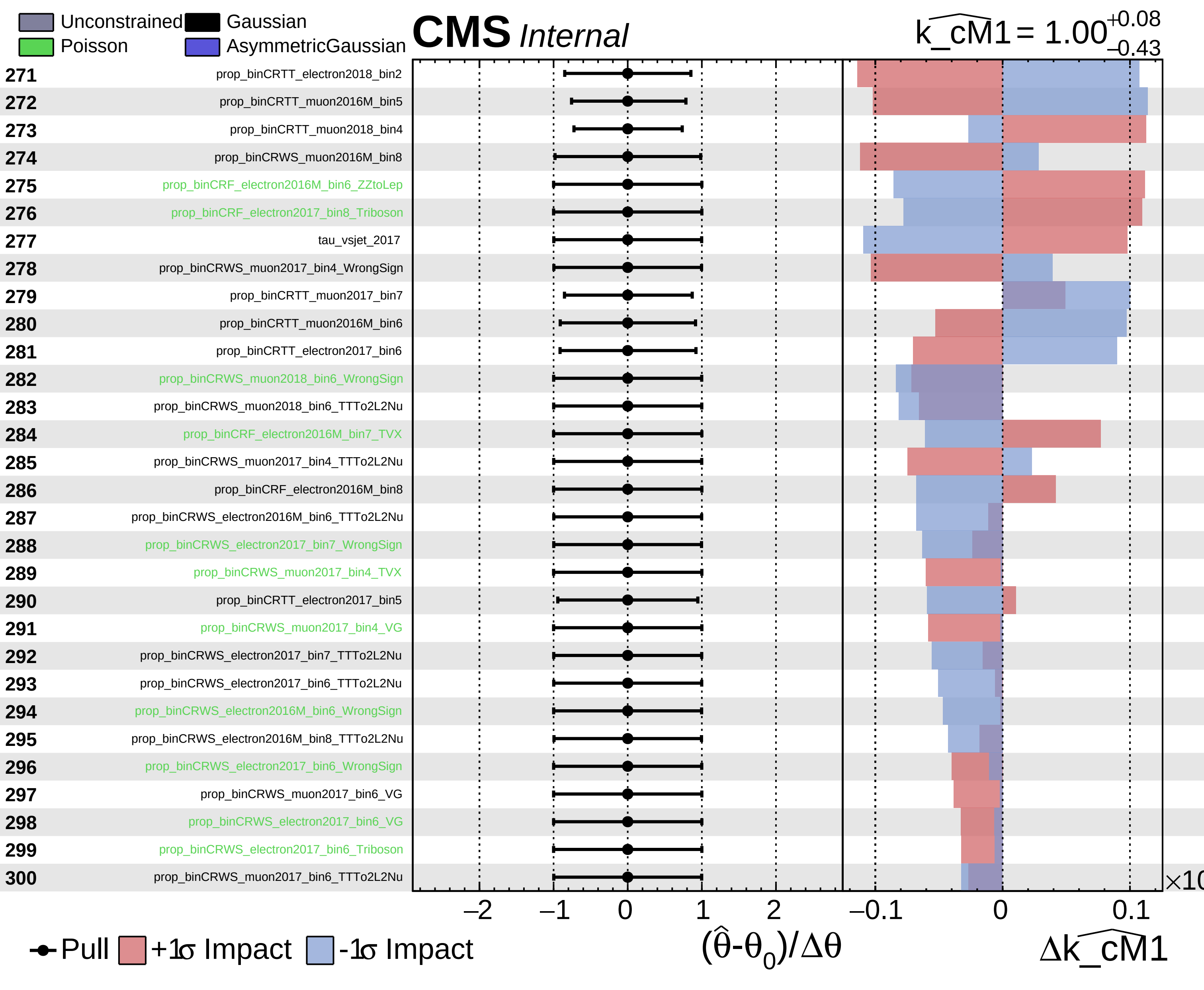

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
181 prop_binCRWS_electron2016M_bin1
182 prop_binCRF_electron2018_bin7
183 prop_binCRWS_muon2017_bin4_Fake
184 prop_binCRF_electron2017_bin4
185 prop_binCRF_muon2017_bin8_TVX
186 puID
187 prop_binCRTT_muon2016M_bin2
188 prop_binCRTT_muon2017_bin3
189 QCDScale
190 prop_binCRTT_electron2017_bin3
191 prop_binCRF_electron2018_bin5
192 prop_binCRTT_muon2017_bin1
193 prop_binCRTT_muon2016M_bin8
194 prop_binCRTT_muon2018_bin8
195 prop_binCRTT_electron2018_bin4
196 prop_binCRF_electron2017_bin8_TVX
197 prop_binCRTT_electron2017_bin1
198 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
199 prop_binCRF_electron2018_bin8
200 prop_binCRTT_electron2016M_bin0
201 prop_binCRTT_muon2016M_bin4
202 prop_binCRWS_electron2018_bin7
203 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
204 prop_binCRTT_muon2018_bin5
205 TES
206 prop_binCRWS_muon2016M_bin5
207 prop_binCRWS_electron2016M_bin8_WrongSign
208 prop_binCRF_muon2018_bin7
209 prop_binCRTT_muon2016M_bin7
210 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1

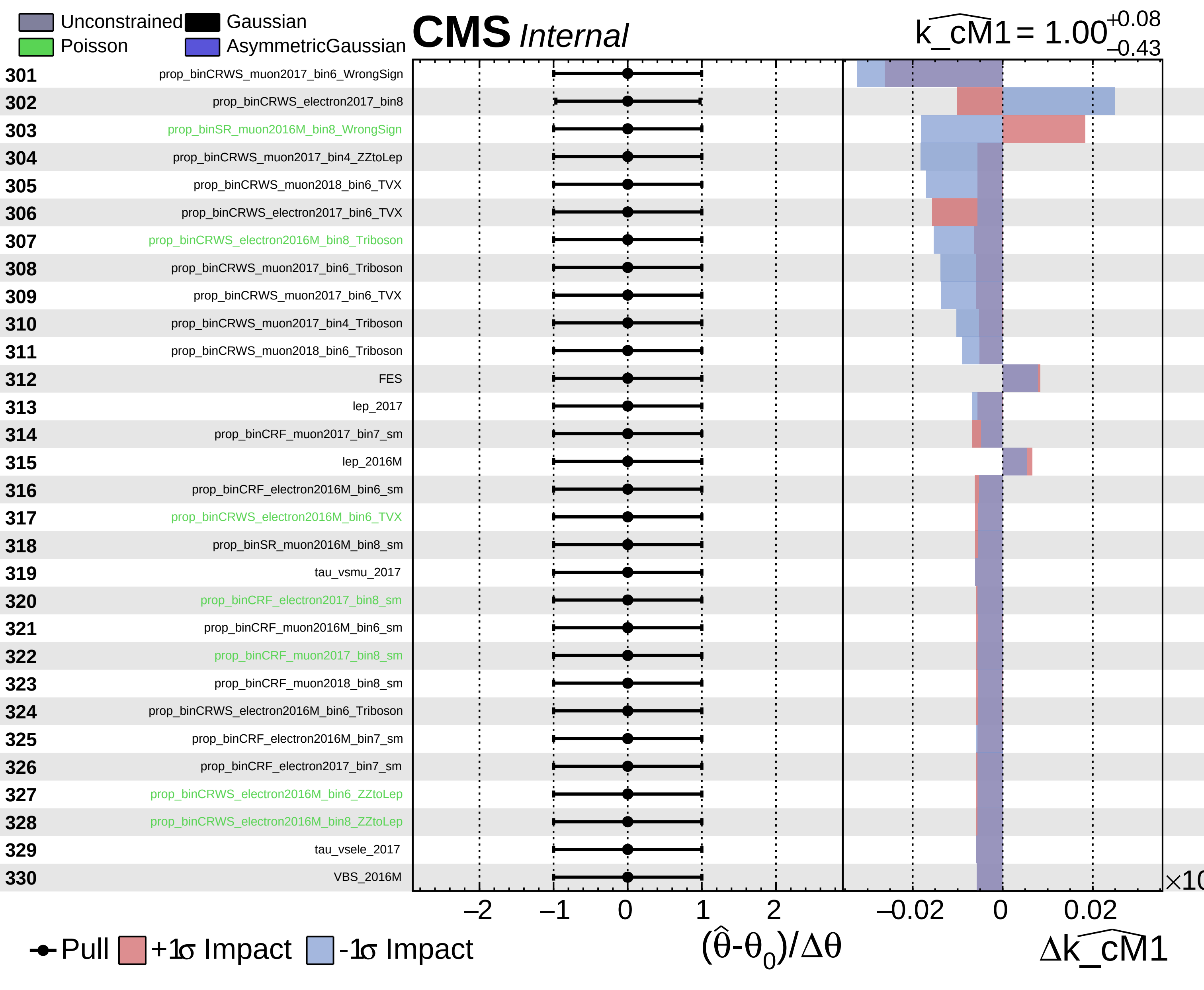


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
211 prop_binCRTT_muon2017_bin8
212 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
213 prop_binCRF_muon2017_bin7_WrongSign
214 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
215 prop_binCRTT_electron2018_bin3
216 prop_binCRWS_electron2016M_bin7
217 prop_binCRWS_muon2017_bin5
218 prop_binCRTT_muon2016M_bin1
219 prop_binCRWS_electron2018_bin6
220 prop_binCRF_muon2017_bin7_Triboson
221 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
222 prop_binCRF_muon2017_bin6
223 prop_binCRWS_muon2017_bin6_Fake
224 prop_binCRF_electron2017_bin7_WrongSign
225 prop_binCRF_muon2016M_bin6_WrongSign
226 prop_binCRTT_muon2017_bin2
227 prop_binCRTT_electron2016M_bin5
228 prop_binCRTT_muon2017_bin5
229 lumi_2018
230 prop_binCRTT_muon2018_bin1
231 prop_binCRWS_muon2018_bin7
232 prop_binCRWS_muon2016M_bin7
233 prop_binCRF_electron2017_bin7_Triboson
234 prop_binCRTT_muon2018_bin3
235 prop_binCRTT_electron2018_bin6
236 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
237 prop_binCRF_electron2017_bin7_TVX
238 prop_binCRWS_muon2017_bin7
239 prop_binCRWS_muon2018_bin8
240 prop_binCRF_muon2017_bin7_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM1

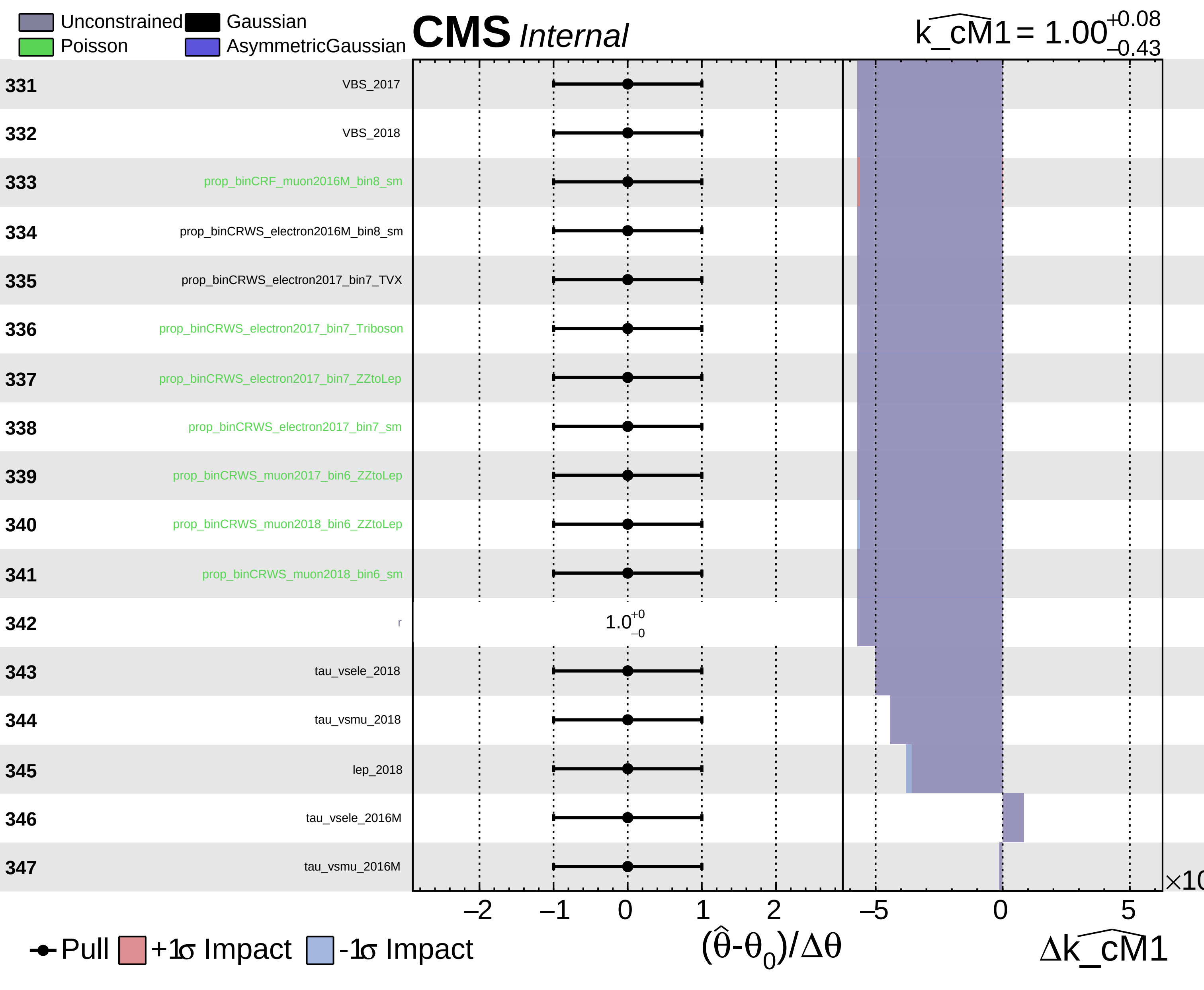


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 1.00^{+0.08}_{-0.43}$
241 prop_binCRTT_electron2017_bin2
242 prop_binCRWS_electron2018_bin8
243 prop_binCRTT_muon2017_bin4
244 prop_binCRTT_muon2016M_bin3
245 lumi_2016M
246 prop_binCRTT_electron2016M_bin3
247 prop_binCRTT_electron2016M_bin8
248 prop_binCRTT_electron2016M_bin4
249 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
250 prop_binCRTT_electron2018_bin8
251 prop_binCRWS_muon2018_bin6_Fake
252 prop_binCRF_muon2016M_bin6_Triboson
253 prop_binCRWS_electron2017_bin6_Fake
254 prop_binCRTT_electron2016M_bin7
255 tau_vsjet_2018
256 prop_binCRTT_electron2016M_bin2
257 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
258 prop_binCRTT_electron2018_bin1
259 prop_binCRWS_electron2016M_bin6_Fake
260 prop_binCRTT_muon2017_bin6
261 lumi_2017
262 prop_binCRTT_electron2017_bin4
263 prop_binCRTT_electron2016M_bin6
264 prop_binCRTT_electron2017_bin7
265 prop_binCRWS_muon2017_bin8
266 prop_binCRTT_electron2018_bin7
267 prop_binCRF_muon2018_bin8_Triboson
268 prop_binCRTT_electron2016M_bin1
269 prop_binCRTT_electron2018_bin5
270 prop_binCRWS_electron2017_bin7_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM1} = 1.00^{+0.08}_{-0.43}$
271 prop_binCRTT_electron2018_bin2
272 prop_binCRTT_muon2016M_bin5
273 prop_binCRTT_muon2018_bin4
274 prop_binCRWS_muon2016M_bin8
275 prop_binCRF_electron2016M_bin6_ZZtoLep
276 prop_binCRF_electron2017_bin8_Triboson
277 tau_vsjet_2017
278 prop_binCRWS_muon2017_bin4_WrongSign
279 prop_binCRTT_muon2017_bin7
280 prop_binCRTT_muon2016M_bin6
281 prop_binCRTT_electron2017_bin6
282 prop_binCRWS_muon2018_bin6_WrongSign
283 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
284 prop_binCRF_electron2016M_bin7_TVX
285 prop_binCRWS_muon2017_bin4_TTTo2L2Nu
286 prop_binCRF_electron2016M_bin8
287 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
288 prop_binCRWS_electron2017_bin7_WrongSign
289 prop_binCRWS_muon2017_bin4_TVX
290 prop_binCRTT_electron2017_bin5
291 prop_binCRWS_muon2017_bin4_VG
292 prop_binCRWS_electron2017_bin7_TTTo2L2Nu
293 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
294 prop_binCRWS_electron2016M_bin6_WrongSign
295 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
296 prop_binCRWS_electron2017_bin6_WrongSign
297 prop_binCRWS_muon2017_bin6_VG
298 prop_binCRWS_electron2017_bin6_VG
299 prop_binCRWS_electron2017_bin6_Triboson
300 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
301 prop_binCRWS_muon2017_bin6_WrongSign
302 prop_binCRWS_electron2017_bin8
303 prop_binSR_muon2016M_bin8_WrongSign
304 prop_binCRWS_muon2017_bin4_ZZtoLep
305 prop_binCRWS_muon2018_bin6_TVX
306 prop_binCRWS_electron2017_bin6_TVX
307 prop_binCRWS_electron2016M_bin8_Triboson
308 prop_binCRWS_muon2017_bin6_Triboson
309 prop_binCRWS_muon2017_bin6_TVX
310 prop_binCRWS_muon2017_bin4_Triboson
311 prop_binCRWS_muon2018_bin6_Triboson
312 FES
313 lep_2017
314 prop_binCRF_muon2017_bin7_sm
315 lep_2016M
316 prop_binCRF_electron2016M_bin6_sm
317 prop_binCRWS_electron2016M_bin6_TVX
318 prop_binSR_muon2016M_bin8_sm
319 tau_vsmu_2017
320 prop_binCRF_electron2017_bin8_sm
321 prop_binCRF_muon2016M_bin6_sm
322 prop_binCRF_muon2017_bin8_sm
323 prop_binCRF_muon2018_bin8_sm
324 prop_binCRWS_electron2016M_bin6_Triboson
325 prop_binCRF_electron2016M_bin7_sm
326 prop_binCRF_electron2017_bin7_sm
327 prop_binCRWS_electron2016M_bin6_ZZtoLep
328 prop_binCRWS_electron2016M_bin8_ZZtoLep
329 tau_vsele_2017
330 VBS_2016M
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM1 = 1.00 +0.08 −0.43
331 VBS_2017
332 VBS_2018
333 prop_binCRF_muon2016M_bin8_sm
334 prop_binCRWS_electron2016M_bin8_sm
335 prop_binCRWS_electron2017_bin7_TVX
336 prop_binCRWS_electron2017_bin7_Triboson
337 prop_binCRWS_electron2017_bin7_ZZtoLep
338 prop_binCRWS_electron2017_bin7_sm
339 prop_binCRWS_muon2017_bin6_ZZtoLep
340 prop_binCRWS_muon2018_bin6_ZZtoLep
341 prop_binCRWS_muon2018_bin6_sm
342 r
1.0 +0 −0
343 tau_vsele_2018
344 tau_vsmu_2018
345 lep_2018
346 tau_vsele_2016M
347 tau_vsmu_2016M
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM1